

BULLHORN COULEE RESERVOIR, AB, Canada

WMO: 718390

Lat: 49.295N

Lon: 113.359W

Elev: 3658

StdP: 12.85

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -20.2	(c) -13.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 26.7	(c) 86.3	(d) N/A	(e) 82.8	(f) N/A	(g) 79.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	DB	-30.9	91.7	7.0	4.0	-35.9	94.5	-40.0	96.9	-43.9	99.1	-48.9	102.0	(4)
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	22.9	21.8	31.4	39.8	49.8	57.4	63.7	62.5	54.1	42.2	31.1	19.8
	DBStd	18.92	15.11	15.79	12.61	9.17	7.67	5.77	5.43	5.63	8.05	9.95	13.35	15.88
	HDD50	4465	841	791	579	318	99	8	0	2	50	271	570	936
	HDD65	8680	1306	1211	1043	755	472	237	88	113	331	706	1017	1401
	CDD50	1358	0	0	1	13	94	229	425	390	173	30	3	0
	CDD65	98	0	0	0	0	2	8	48	36	4	0	0	0
	CDH74	2330	0	0	0	7	56	179	931	873	276	8	0	0
	CDH80	634	0	0	0	0	3	32	283	260	55	1	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	21.6	1.1	0.9	1.3	1.9	3.0	4.2	1.9	1.8	1.9	1.3	1.2	1.1
	PrecMax	33.2	1.7	1.9	2.5	3.7	6.5	12.7	6.8	4.6	5.3	3.1	2.4	2.3
	PrecMin	13.5	0.3	0.2	0.5	0.7	0.4	0.9	0.1	0.1	0.2	0.2	0.4	0.4
	PrecStd	4.7	0.4	0.4	0.5	0.7	1.5	2.6	1.5	1.1	1.1	0.7	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	54.3	59.1	73.4	79.4	85.2	91.4	90.6	86.0	72.5	63.5	51.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	46.7	48.9	54.4	65.7	75.1	78.4	86.7	86.0	80.9	66.8	56.2	46.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	43.7	44.8	51.3	60.2	70.9	75.2	82.9	82.7	77.0	61.9	51.6	42.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	41.3	41.4	47.6	55.3	66.0	71.6	79.3	79.1	71.7	57.1	47.7	39.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	18.0	19.8	19.6	20.7	21.7	22.2	26.7	28.1	25.5	21.1	18.5	18.5
		MCDBR	16.9	19.7	23.0	30.4	30.4	29.1	33.6	36.5	37.2	27.9	21.9	18.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.245	0.259	0.279	0.328	0.336	0.330	0.335	0.352	0.314	0.281	0.257	0.231	
	taud	2.409	2.407	2.445	2.338	2.394	2.456	2.448	2.386	2.466	2.542	2.495	2.423	
	Ebn at Noon	266	288	299	292	292	294	291	281	282	277	261	258	
	Edh at Noon	20	25	29	36	36	34	34	34	28	22	18	17	
All-Sky Solar Radiation	RadAvg	394	712	1103	1524	1816	2030	2254	1798	1278	792	441	315	
	RadStd	33	53	96	101	158	155	123	113	104	78	34	30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page